

Module 06: Learning in High School

Learning Objectives

1. Define **Learning** within the context of High School.
2. Analyze how Learning interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Learning.

Introduction

This module explores **Learning**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Learning is a critical component of the 8-part Active Inference spine, bridging the gap between Action and Communication.

Key Concepts

1. Learning as a Markov Blanket Boundary

How does Learning define the boundary between the agent and the environment?

2. Generative Models of Learning

What parameters involved in Learning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Learning drive the perception-action loop?

Applications

In High School, we see Learning manifest in: * **Specific Example 1:** When Spotify's Discover Weekly gets better at recommending songs you like over months of use, it is performing

Active Inference learning: each song you skip or save generates a prediction error that updates the algorithm's generative model of your music taste, gradually reducing the surprise in its weekly recommendations until the playlist feels like it reads your mind. * **Specific Example 2:** Training an image classifier to recognize cats vs. dogs illustrates machine learning as Active Inference: the neural network starts with random weights (high free energy, massive prediction errors on training images), and through thousands of backpropagation iterations it adjusts its generative model parameters until the prediction error on new images is minimized -- the loss curve dropping over epochs is literally a plot of free energy being minimized.

Conclusion

Understanding Learning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.